GAYATRI VIDYA PARISHAD COLLEGE OF ENGINEERING FOR WOMEN
(Autonomous)
(Approved by AICTE, New Delhi and Permanently Affiliated to Andhra University, Visakhapatnam)
Madhurawada :: Visakhapatnam – 530 048

ANNEXURE-XLVI

MICROPROCESSOR & CONTROLLERS WITH INTERFACING

(Open Elective)

Course Code: 24EC11EL04 **L-T-P-C: 3-0-0-3**

Course Outcomes: At the end of the Course, the student shall be able to	
CO1:	Comprehend the architecture and working of 16 bit microprocessor 8086.
CO2:	Apply assembly language programming skills to perform arithmetic, logical and string operations with 8086.
CO3:	Develop applications involving interfacing of various peripherals with 8086 microprocessor.
CO4:	Analyze the architecture and working of 8-bit microcontroller 8051
CO5:	Develop 8051 microcontroller based standalone applications for societal needs.

UNIT-I	**09 Lectures**
Introduction: Difference between Microprocessor and Microcontroller, Features of a Microprocessor, Von Neumann and Harvard Architecture, RISC and CISC Instruction Set **8086 Microprocessor:** Main features, internal architecture, register organization, pin diagram/ description, Memory Organization, Deriving the System Bus, minimum mode and maximum mode configuration	

UNIT-II	**09 Lectures**
8086 Programming: Addressing Modes, Instruction Set, Assembler Directives, simple programs with an assembler, interrupts and interrupt response	

UNIT-III	**12 Lectures**
8086 Interfacing: Semiconductor memory interfacing, Intel 8255 programmable peripheral interface Interfacing switches and LEDS, Interfacing seven segment displays, Interfacing Stepper Motor. Interfacing 8251 USART, Interfacing DMA Controller, Interfacing 8259 PIC	

UNIT-IV	**09 Lectures**
8051 Architecture: Introduction to microcontrollers, Architecture, Register Organization, 8051 pin description, basic connections, I/O ports, memory organization, Interrupt Control, Power Control, Timers/Counters, Serial Communication.	

UNIT-V	**09 Lectures**
8051 Programming and Interfacing: Addressing modes, Instruction Set, Programs using Interrupts, Timers and Serial Communication, Interfacing switches and LEDS, Interfacing seven segment displays, Interfacing Stepper Motor.	

Text Books:
1. A.K Ray, K.M.Bhurchandhi," Advanced Microprocessor and Peripherals", Tata McGraw Hill Publications, 2000 2. The 8051 Microcontrollers and Embedded systems Using Assembly and C, Muhammad Ali Mazidi and Janice Gillespie Mazidi and Rollin D.McKinlay; Pearson 2-Edition, 2011
Reference Books:
1. Microprocessors and Interfacing – Programming and Hardware by Douglas V Hall, SSSP Rao, Tata McGraw Hill Education Private Limited, 3rd Edition, 2017 **2.** The 8051 Microcontroller & Embedded Systems Using Assembly and C by Kenneth J.Ayala, Dhananjay V.Gadre, Cengage Learning , 1st edition, 2010.
Web References:
1. https://onlinecourses.nptel.ac.in/noc20_ee11/preview (Microprocessors and Interfacing) 2. https://onlinecourses.nptel.ac.in/noc22_ee12/preview (Microprocessors and Microcontrollers)